

Appendix: Mathematical Details for the 2D Supersonic Marshak-Wave Model

Yuval Kochavi et al.

July 12, 2026

Purpose

This appendix collects the full mathematical expressions and intermediate derivation steps that are only cited in the PRL main text due to the five-page limit.

1 1D Hammer–Rosen Backbone

Using power-law opacity and material energy,

$$\frac{1}{\kappa} = g T^\alpha \rho^{-\lambda}, \quad e = f T^\beta \rho^{-\mu}, \quad (1)$$

the front position is written as

$$x_F^2(t) = \frac{2 + \varepsilon}{1 - \varepsilon} C H^{-\varepsilon}(t) \int_0^t H(t') dt', \quad (2)$$

with

$$H(t) = T_s^{4+\alpha}(t), \quad \varepsilon = \frac{\beta}{4 + \alpha}, \quad C = \frac{16 g \sigma_{\text{SB}}}{3 f (4 + \alpha)} \rho^{2-\mu-\lambda}. \quad (3)$$

The coupled boundary-flux closure uses the Marshak boundary condition and material-energy balance:

$$\sigma_{\text{SB}} T_D^4 = \sigma_{\text{SB}} T_s^4 + \frac{1}{2} F(0, t), \quad \dot{E}(t) = F(0, t). \quad (4)$$

2 2D Wall-Energy Loss

For a cylindrical tube of radius R , the cumulative wall-deposited energy is

$$E_W(t) = 2\pi R \int_0^{x_F(t)} \int_0^t \mathcal{H}(t - t_0(x)) \dot{E}_{\text{mat}}(t') dt' dx. \quad (5)$$

For gold sleeves, the wall-heating fit used in the model is

$$E_{\text{gold}}(t) = 0.59 T_0^{3.35} (t - t_0(x))^{0.59} \quad (\text{hJ/mm}^2). \quad (6)$$

Material-specific coefficients for Be and Cu are taken in the same form from the cited wall-heating literature.

3 Wall Ablation and Reduced Input Area

For gold, the subsonic self-similar ablation velocity is

$$u_{\text{gold}}(t) = -510.1 \tilde{u}(\rho_{\text{gold}}) T_0^{0.716} (t - t_0(x))^{0.036} \quad (\text{km/s}). \quad (7)$$

The local radius update is

$$R(t, x) = R_0 - u_W(t) (t - t_0(x)), \quad (8)$$

which modifies incoming energy as

$$E_{\text{in}}(t) = 2\pi \int_0^t R_0^2 \left(1 - \frac{u_W(t')t'}{R_0^2}\right)^2 F(0, t') dt'. \quad (9)$$

Ablation-driven compression updates the mean density through

$$\rho(t) = \rho_0 \frac{V_0}{V(t)}, \quad V(t) = \int_0^{x_F(t)} \pi R^2(t, x) dx, \quad (10)$$

and therefore makes C time dependent:

$$C(t) = \frac{16 g \sigma_{\text{SB}}}{(4 + \alpha) 3 f t} \int_0^t \rho^{\mu+\lambda-2}(t') dt'. \quad (11)$$

4 Finite-Diameter Correction

When λ_R approaches or exceeds $2R$, we use

$$\lambda_{\text{eff}} = [(2R)^{-n} + \lambda_R^{-n}]^{-1/n}, \quad (12)$$

so the diffusion coefficient becomes $D = c\lambda_{\text{eff}}/3$.

5 Cylindrical Bent-Front Reconstruction

The leading cylindrical bent-wave representation is

$$x_F(r, t) = \frac{1}{\kappa_0} \cosh^{-1} \left(1 + \frac{D\kappa_0^2 t}{2}\right) J_0(\kappa_0 r) + \text{H.O.T.}, \quad (13)$$

with κ_0 obtained from the wall-loss eigenvalue problem (or the high-albedo approximation where applicable).

6 Time-Dependent Albedo Closure

Albedo is evaluated from the wall energy balance:

$$\alpha(t) = \frac{\sigma_{\text{SB}} T_{s,\text{int}}^4 - \frac{1}{2} \frac{dE_{\text{wall}}}{dt}}{\sigma_{\text{SB}} T_{s,\text{int}}^4 + \frac{1}{2} \frac{dE_{\text{wall}}}{dt}}. \quad (14)$$

Then

$$\varepsilon(t) = \frac{3}{4} \rho \kappa R [1 - \alpha(t)], \quad (15)$$

which sets the curvature strength through $\kappa_0(t)$.

7 2D Temperature and Observable Flux

Given $z_F(r, t)$, we reconstruct

$$T(r, z, t) = T_s(t) \left(1 - \frac{z}{z_F(r, t)} \right)^{\frac{1}{4+\alpha-\beta}}, \quad (16)$$

and compute

$$\Phi(r, z_{\text{det}}, t) = \sigma_{\text{SB}} T^4(r, z_{\text{det}}, t) \quad (17)$$

for comparison with diagnostic emission profiles.

Source Mapping

The wall-loss, wall-heating, and ablation formulas above are taken from Refs. Cohen et al. (2018, 2020). The bent-wave framework is from Hurricane and Hammer (2006), with cylindrical modeling details and modern comparisons from Courtois et al. (2022, 2024). The 1D HR backbone follows Hammer and Rosen (2003) and Rosen (2009).